Open System-One Decision Models: Frozen Diffusion-LM Backbones and Specialist Decision Heads for Calibrated Scientific Decisions

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Abstract

Structured decisions—selecting among options, verifying whether evidence supports a claim, and grading response quality on ordinal scales—are the atomic operations of automated scientific pipelines. Proprietary "System-One" models deliver such decisions as typed API calls without generation, but are closed. We present an open, trainable counterpart: a frozen masked-diffusion language model backbone (LLaDA-8B) with small specialist decision heads producing probabilities for three decision types (choice, noul, score). We contribute three findings. (1) A deterministic canonical ordering gives a permutation-invariant input sequence for options with distinct token representations, rather than relying on train-time order augmentation. (2) Multi-layer attention pooling over option spans outperforms last-layer mean pooling for choice (+7 points), and the combined specialist, canonical, and ordinal recipe improves graded scoring (+9 points). (3) Scaling training data 30× on an out-of-domain corpus improves in-distribution accuracy to 0.96 but does not transfer (−13 points on the target domain), suggesting that volume alone is insufficient. On public benchmarks, the release heads reach 0.8500 on SciFact claim verification under the legacy conversion but 0.464 under the canonical `systemone-v2` encoding—an encoding sensitivity of the release heads—while evidence-corruption retraining removes the option-side shortcut and recovers 0.693. The heads transfer zero-shot to general classification (0.9300 SST-2, 0.8537 AG News); knowledge-intensive GPQA remains difficult (0.3125 on main). Candidates using LoRA domain adaptation score higher on elite choice but lower on GPQA and SciFact. Their head-training schedules also differ, so this is a candidate comparison, not an isolated causal test of DAPT. The release retains the frozen-backbone heads.

1 Introduction

Automated scientific pipelines—agentic literature review, claim verification, data extraction, routing to slower reasoning components—are built out of small structured judgments: *which* of these options answers the question, *does* the evidence support the claim, *how* good is this candidate answer. Today these judgments are typically extracted from generative LLMs by prompting for JSON and parsing the output, a pipeline that is slow, brittle under format drift, and poorly calibrated.

A recent class of models removes generation entirely: given an unstructured state and a set of typed questions, a *System-One* model returns typed values—a choice among options, a yes/no probability, an ordinal score—each with a calibrated probability, in a single forward pass. The

flagship proprietary model of this class is Jev (TypeSafe AI, 2026); open reimplementations appeared within days of its release (kev, openjev, laya, SemIf), all built on autoregressive or encoder backbones.

We ask what an open System-One model needs to be good at, and build one on the other major family of language models: a *masked-diffusion* backbone (LLaDA-8B), whose bidirectional attention is a natural fit for decision readouts that compare each option against the full context. To preserve the pretrained weights and limit trainable parameters, we train specialist heads per decision type on frozen multi-layer hidden states. Our contributions:

- **Permutation invariance by canonical ordering.** Sorting options deterministically by content makes the presentation order a function of the option set. For distinct token representations, permutations produce the same input sequence; train-time order averaging alone does not impose this constraint.
- **Specialist heads per decision type.** Multi-layer attention pooling over option spans improves choice (+7 points over mean pooling) while hurting binary verification—different decision types want different readouts. The combined specialist, canonical, and ordinal recipe improves graded scoring (+9 points).
- **Data volume and adaptation are not sufficient.** A 30× out-of-domain data increase fails to transfer (−13 points). We also compare release candidates with and without LoRA domain adaptation. The adapted candidates improve some elite metrics but lose accuracy on GPQA and SciFact. We retain the frozen-backbone candidate; matching the head optimization schedules is necessary before drawing causal conclusions about adaptation.

We evaluate on the elite scientific battery, GPQA, SciFact, and four classification datasets. Metrics cover accuracy, calibration, permutation consistency, and retrospective automation at 5% error.

2 Setup

Decision types. We consider three typed decisions over a state (context) and a question: **choice** selects one of K answer options; **noul** decides whether evidence supports a claim (K=2); **score** assigns an ordinal grade $0 < 1 < 2$ (unrelated / related-but-wrong / correct).

Model. The released span-pooling heads encode context, question, and options as one observed token sequence, without inserting mask markers or generating text. A frozen masked-diffusion backbone (LLaDA-8B-Instruct) produces token representations. A mean-pooled MLP reads final-layer option spans for noul and score; *AttnPool* learns a query for each of layers $\{8, 16, 24, 32\}$ and concatenates the pooled features for choice. The released choice head has 67,166,209 parameters; each MLP has 16,793,601. Only these heads are trained during decision training.

Canonical ordering. Options are sorted by token-id content before encoding, and gold indices remapped. For options that remain distinct after tokenization and truncation, any input permutation collapses to the same sequence. Identical or truncated-to-identical options require a separate tie policy if their external identifiers must remain distinguishable.

Ordinal auxiliary loss. For score we add a CORAL-style term $\sum_{j=1}^{K-1} \mathrm{BCE}\big(P(y \geq j),\, \mathbf{1}[g \geq j]\big)$ on cumulative softmax mass, respecting the grade order without changing the K-way decode.

Metrics. We report accuracy, expected calibration error (ECE), option-order flip rate (a change in the selected option's original identity after permutation), and *automation@5%err*: the largest confidence-ranked prefix with realized error $\leq 5\%$. Automation is retrospective on labeled evaluation data, not a guarantee of deployment error. Historical non-canonical flip figures compared output positions without remapping option identities; they are excluded here pending re-evaluation. The release fixes that metric and includes regression tests.

3 Experiments

3.1 Scientific decision battery

The *elite* battery is built from passage-grounded scientific QA typed by reasoning kind (factual, causal, multihop, negation, numerical, definitional); each passage generates a choice item (K=4, one gold + three distractors), a noul item (supported vs. unsupported claim), and a score item (correct / related-but-wrong / unrelated answer). Splits are passage-disjoint to prevent leakage (2,240/6,720/6,720 train rows per type). The *v2* battery scales the same construction to 25,000 passages from three raw QA corpora (physics/polymer LaTeX full texts, astronomy, and a derived set) with groundedness filters—content-word recall ≥ 0.5 and numeric binding—yielding ∼470,000 training decisions; we use it to probe the data-scaling question.

3.2 Public benchmarks

GPQA main (448 graduate-level questions) and diamond subset (198) as choice; SciFact dev (340 claim–abstract pairs) as noul and 3-way score; plus general classification (SST-2, AG News, Enron spam, Banking77) to probe out-of-domain transfer. Benchmark examples are not used for gradient updates or temperature fitting; temperatures are fitted on the battery dev split. Benchmark outcomes did inform the choice of release candidate.

3.3 Candidate protocols and comparison limits

The frozen release heads (`c_*`) use learning rate 3×10^{-4}, accumulation 1, warmup 200, and 2,000/3,000/3,000 example steps for choice/noul/score. Heads for the adapted candidates (`dag_*`, `da_*`) and the small-backbone control (`eb_*`) instead use learning rate 10^{-3}, accumulation 8, warmup 100, and 3,000 steps per type. All use seed 7331, the same elite training files, the same head architecture per type, and an ordinal auxiliary weight of 1 for score. Thus these are candidate comparisons, not a controlled DAPT-only ablation.

DAPT uses LoRA rank 16, alpha 32, dropout 0.05, and the masked-token objective implemented in `sciev.dapt`. With batch size 16 and sequence cap 1,024, the 2,000/5,000-step adapters correspond to at most 32.8/81.9 million input token slots, not measured counts of unique non-padding tokens. The 5,000-step run warm-starts from the 2,000-step adapter with a fresh optimizer and a schedule recomputed for the shorter total budget; optimizer and RNG state were not restored. The 1.4B `bw1_sr` control is checkpoint-g20 from a 20-step run, with a 768-token window versus 2,048 for LLaDA. It is an early-training control, not evidence about the capacity of a fully trained 1.4B model.

	choice	noul	score
Baseline (mixed MLP)	0.775	0.707	0.522
+ orders= 4	0.721	0.782	0.586
+ AttnPool	0.847	0.636	0.588
+ both	0.749	0.780	0.576
Per-type heads	0.876	0.710	0.574
Per-type + canon. + ord.	**0.870**	**0.783**	**0.608**

Table 1: Accuracy ablations on the scientific battery (frozen LLaDA-8B). The release combines specialist heads, canonical ordering, and ordinal scoring. Historical, unaligned positional flip figures are not reported.

4 Results

4.1 Ablations on the scientific battery

Key findings: train-time logit averaging costs choice accuracy in these runs, while canonical ordering deterministically fixes presentation order. AttnPool on intermediate layers helps choice (+7) but hurts noul, motivating per-type heads. The final combined recipe lifts score from 0.522 to 0.608; this comparison alone does not isolate the ordinal loss from the other changes.

4.2 Data scaling does not transfer across domains

	in-distribution (v2)	elite (target)
choice	0.965	0.738
noul	0.942	0.782
score	–	0.616

Table 2: Heads retrained on 30× more data from a different QA distribution (qa_raw corpora) reach near-ceiling in-distribution accuracy but do not transfer to the elite battery, whose questions stress causal/multihop/negation reasoning absent from the source corpus.

4.3 Public benchmarks

4.4 Backbone-only GPQA diagnostics

Low GPQA accuracy could reflect either a weak decision readout or missing backbone knowledge. We scored the cleaned GPQA cohort with the frozen backbone alone, bypassing every trained head: three pseudo-log-likelihood variants over the option text (full-mask, true leave-one-out, and single-token label scoring) and two greedy masked-diffusion generation modes (direct 6-token answer and 64-token reasoning). Table 4 shows that no head-free protocol exceeds chance by much: the best backbone-only readout (label scoring, 0.284/0.301) sits at or below the trained head (0.295/0.307 in the matched systemone-v2 study), and 64-token reasoning generation returns to chance. A dedicated choice head trained on our held-out general-MCQA battery likewise leaves GPQA at 0.27 while reaching 0.84 on its own evaluation split. The consistent picture is that the backbone, not the readout, limits GPQA: the knowledge required by graduate-level, retrieval-resistant questions is not present in extractable form. We therefore treat GPQA accuracy as a backbone capability ceiling

benchmark	type	n	accuracy	ECE	auto@5%
GPQA main	choice	448	0.3125	0.3180	0.0112
GPQA diamond	choice	198	0.3182	0.3089	0.0152
SciFact dev	noul	340	0.8500	0.0998	0.2706
SciFact dev	score	340	0.6294	0.0796	0.2559
Enron spam	choice	2000	0.7510	0.1046	0.0000
SST-2	choice	872	0.9300	0.0682	0.9415
AG News	choice	7600	0.8537	0.0556	0.3674
Banking77 (K=77)	choice	3080	0.2289	0.0809	0.0195

Table 3: Public-benchmark results for the frozen `c_*` release heads, transcribed from the archived evaluation JSONs rather than earlier rounded summaries. Benchmark data were not used for gradient updates or temperature fitting. Temperatures fitted on elite dev were 0.8274, 2.7127, and 0.9367 for choice, noul, and score. Counts and metrics describe our converted evaluation subsets; third-party results with unverified protocol equivalence are not included as directly comparable rows. The SciFact rows use the legacy single-token conversion; the canonical-encoding comparison and an anti-leakage variant are in Table 10. Machine-readable values and source hashes are in `paper/results.json`.

rather than a decision-layer defect, and the remaining lever is grounding — supplying evidence in the state — not further head optimization.

method	main (n=440)	diamond (n=193)
full-mask PLL	0.225	0.207
leave-one-out PLL	0.234	0.249
label scoring	0.284	0.301
prompted generation (6 tok, direct)	0.259	0.249
prompted generation (64 tok, reasoning)	0.250	0.228
release choice head (reference)	0.295	0.307
qa-v2 choice head (reference)	0.273	0.269

Table 4: Backbone-only GPQA accuracy (no trained head), Phase-A diagnostics on the cleaned cohort. PLL methods score the option text directly; prompted modes generate an answer via masked diffusion. Reference rows are trained heads evaluated on the same cohort. Random chance is 0.25. No backbone-only method meaningfully exceeds the trained head, indicating a backbone knowledge ceiling rather than a readout bottleneck.

4.5 Contrastive bi-encoder control

To test whether the joint cross-encoding is itself the active ingredient, we trained a CLM-style contrastive arm on the same frozen backbone and the same training cohort: state/action projection heads over mean-pooled backbone vectors, bidirectional InfoNCE with same-question and in-batch negatives, protocol-identical held-out temperature and acceptance policy. At our data scale the dot product is far behind (Table 5): 55–65 points down on the in-domain suites and at chance on GPQA. The finding is consistent with CLM-8B's own recipe, which relies on roughly 10^7–10^8 contrastive pairs plus synthetic hard negatives; at $\sim 5\times10^4$ decisions, an option that never attends to the state cannot recover the interaction. The arm is a single configuration, so we claim a matched-scale

comparison rather than a verdict on bi-encoders in general. A natural follow-up use — reusing the cached embeddings as a cheap dot-product shortlist over decisions with more than four options — is also ruled out at this scale: on the 3080-row, 77-option banking77 suite the gold label reaches the top-4 of the arm's ranking only 8.9% of the time (recall@32: 46.5%), and suite-level action retrieval stays equally weak (recall@64 15–40%). Any K>4 path therefore runs through the joint head itself, e.g. option-chunked tournaments, not embedding prefiltering.

suite	n	joint head	bi-encoder
qa_eval	5021	0.838	0.376
sci_elite	448	0.920	0.263
gpqa_main	440	0.273	0.241
gpqa_diamond	193	0.269	0.244

Table 5: Matched-architecture control: the CLM-style contrastive bi-encoder on our frozen backbone and training cohort versus the joint cross-encoded choice head evaluated on the same files (`runs/sci/eval_bi_v2_s7331`). Selection accuracy on the bi-encoder's dev split plateaued at 0.371. Joint-head numbers are from the same qa-v2 cohort evaluation; GPQA cells remain near chance for both systems, consistent with the backbone ceiling of Table 4.

4.6 Grounded GPQA: evidence rescues the head

The backbone ceiling of Table 4 characterizes *closed-book* GPQA. The system's actual contract, however, is grounded decisions: evidence in the state, typed options. To measure whether the joint readout can exploit evidence at all, we built paired grounded variants of the cleaned GPQA cohort — identical options and ordering, only the state gains an evidence block — in two regimes: *oracle* appends the benchmark's expert-written explanation (a gold-revealing upper bound), and *nev* appends a non-expert validator's own reasoning trace (noisy; validators average roughly a third correct).

Table 6 separates the failure modes cleanly. On GPQA-main, oracle evidence lifts the release head from 0.295 to 0.695 and the qa-v2 head from 0.273 to 0.707: given the right document, the same backbone and head that sit at chance decide correctly on graduate-level items — and both heads converge, so evidence use lives in the joint architecture, not the fine-tune. The noisy nev trace recovers almost nothing (0.318 vs. 0.295), so the oracle–nev gap prices retrieval quality: usable evidence must be explanation-grade, not topical. A real production retrieval stack confirms the price: BM25 top-3 passages from a 30.8M-article PubMed index land at 0.304 on main for the release head—below nev and statistically indistinguishable from closed-book (0.295)—and 0.292 for qa-v2. Coverage is not the bottleneck (only 2/440 questions found no hit); the retrieved abstracts are topically adjacent rather than explanation-grade, and GPQA is by construction resistant to topical evidence. Closing the oracle gap therefore requires retrieval that returns reasoning-bearing text—or a corpus that covers physics and chemistry, which PubMed largely does not. GPQA-diamond resists even the oracle (0.326 vs. 0.307 baseline): on the hardest subset the residual failure is option discrimination, not missing knowledge. Grounding thus converts most of the closed-book ceiling, and what remains on diamond is a harder interaction limit for this backbone.

The counterfactual-trained `alp25x` choice head (Section 4.10) is the first arm to exceed the leaky release on explanation-grade evidence (0.725 versus 0.695 on main oracle) while simultaneously dropping 3–5 points on the noisy nev and retrieved arms—the signature of a head that follows evidence when it answers the question and commits less when it does not. Anti-leakage training

therefore did not merely remove the shortcut; it improved the evidence-conditioned axis the benchmark actually prices. The gain is attributable to the counterfactual labels, not to abstention alone: the uniform-only variants `alpm1` and `alp25p` score 0.682 and 0.686 on the same main-oracle arm, below both the release head and `alp25x`.

	release head		qa-v2 head		`alp25x` head	
evidence arm	main	diamond	main	diamond	main	diamond
closed-book (reference)	0.295	0.307	0.273	0.269	0.275	0.326
nev validator trace	0.318	0.332	0.284	0.269	0.280	0.306
retrieved (PubMed BM25)	0.304	0.340	0.292	0.278	0.276	0.293
oracle explanation	0.695	0.326	0.707	0.244	**0.725**	0.306

Table 6: Paired grounded GPQA evaluation (`runs/sci/gpqa_grounded_v2`): evidence appended to the state, options and ordering byte-identical to the closed-book rows (n=438/440 main / n=191/193 diamond; 2 questions per split had no retrieved hit). Oracle is a gold-revealing ceiling, nev a noisy real-world proxy, and retrieved is BM25 top-3 abstracts from a 30.8M-article PubMed FTS5 index per question. Oracle grounding doubles main accuracy on both heads; neither noisy nor retrieved evidence moves it, and diamond stays near chance, marking the discrimination floor. The `alp25x` column is the counterfactual-trained anti-leakage choice head of Section 4.10: it beats the leaky release on the oracle arm and concedes on the noisy arms—evidence-following rather than prior-driven commitment.

4.7 Backbone and adaptation candidates

	elite			GPQA		SciFact	
candidate	choice	noul	score	main	diamond	noul	score
bw1_sr 1.4B (g20)	0.2802	0.6672	0.3344	0.2656	0.2828	0.5941	0.3000
LLaDA-8B frozen (c_*)	0.8698	**0.7825**	**0.6075**	**0.3125**	**0.3182**	**0.8500**	**0.6294**
DAPT g2000 (dag_*)	0.8790	0.7790	0.5978	0.2812	0.2677	0.8118	0.5147
DAPT g5000 (da_*)	**0.8836**	0.7555	0.5926	0.2589	0.2424	0.7235	0.5000

Table 7: Candidate accuracies from the archived JSON reports. Elite sample sizes are 653/1,959/1,959 for choice/noul/score; public-benchmark sizes match Table 3. The frozen c_* heads are retained for release. Head optimization differs between c_* and the other rows, and bw1_sr is an early-training, shorter-context control. Consequently, this table does not isolate adaptation, model size, or training duration as a cause of the differences.

Relative to c_*, the g5000 candidate gains 1.38 percentage points on elite choice but loses 5.36 on GPQA main and 12.65 on SciFact noul. The g2000 candidate improves retrospective elite noul automation from 0.2777 to 0.4125. Neither meets the release gate: at most one point of elite accuracy loss per type and either improved SciFact or at least three points of GPQA-main improvement. These observations support retaining c_* among the tested candidates, not rejecting DAPT in general.

4.8 Matched frozen/adapted comparison on complete evidence

The candidate screen above confounds adaptation with head optimization and uses silently truncated contexts. The `systemone-v2` battery removes both problems: identical recipe `scientific-v1` (spanpool, canonical order, per-type heads, 3×10^{-4} head LR, warmup 200, accumulation 1, seed 7331), group-disjoint splits, and evidence that either fits whole or is excluded with a recorded reason (8,576 long-context exclusions; median passage 783 tokens).

	elite (complete evidence)			GPQA		SciFact	
arm	choice	noul	score	main	diamond	noul	choice
frozen (`fr`)	**0.970±.017**	**0.925±.006**	**0.859±.075**	**0.295±.014**	**0.307±.012**	0.483±.023	**0.479±.018**
adapted g5000 (`da`)	0.961±.011	0.913±.007	0.779±.109	0.288±.018	0.297±.011	**0.709±.020**	0.444±.062

Table 8: Accuracy (mean ± sd over seeds 7331–7333) of the matched frozen/adapted arms on `systemone-v2` (identical head recipe, splits, and evaluation per seed). Elite n: 448/1,200/1,218; GPQA main/diamond: 441/193; SciFact: 332 per task. Dev-fitted temperatures for seed 7331: choice 3.39/2.17, noul 1.49/2.78, score 1.27/1.76 (fr/da). Adaptation robustly improves out-of-distribution verification (SciFact noul +22.6 points, consistent across seeds, from below the 0.60 majority baseline) at a noisy cost to in-distribution ordinal scoring. This isolates the backbone effect the candidate screen could not, and reverses the sign of the SciFact result observed in the unmatched comparison.

Calibration (dev-fitted temperature, frozen at evaluation) is strong in-distribution under seed 7331: elite ECE is 0.0167/0.0189/0.0118 for choice/noul/score under the frozen arm, versus 0.0164, 0.0163, and 0.0823 under the adapted arm. On SciFact noul the frozen arm sits below its own majority baseline (0.483 mean versus 0.5964), while the adapted arm reaches 0.709 mean accuracy (macro-F1 0.7289, ECE 0.1607 at seed 7331): domain adaptation transfers to an external verification regime it was never trained on. The cost concentrates in rubric scoring, where adaptation drops seed-7331 accuracy to 0.7521 (argmax MAE 0.2479 versus 0.1043) with worse calibration—though the three-seed spread (±0.11) shows this gap fluctuates across runs. The matched design therefore rejects a uniform verdict in either direction: adaptation effects are task-dependent, not monotone, and the verification gain is the most robust difference observed.

4.9 Evidence controls expose option-side leakage

High accuracy alone does not prove that decisions condition on the supplied evidence. We therefore evaluate *evidence controls*: empty (the evidence span removed) and shuffle (the span replaced by a donor passage from a different source group in the same split). Controls keep the original question and proposed answer and are scored for *agreement with the reference answer*—not relabelled, since altered evidence has no valid ground truth. Agreement far above chance under destroyed evidence means the model decides without reading the evidence, whether through parametric recall or option-side cues.

The controls qualify the headline accuracies. Choice agreement under empty evidence is 0.87 (frozen): the task is largely solvable without the passage, because gold answers are teacher-written text that matches question phrasing while distractors are numeric perturbations or topically foreign answers. Noul falls to its majority prior under shuffled evidence, as it should. Score behaves as designed—agreement collapses toward chance under wrong evidence—and is the only task whose accuracy directly reflects evidence use. Both arms rely on option-side leakage similarly; adaptation does not fix it. The matched elite numbers should therefore be read as upper bounds on evidence-

arm	kind	original acc	empty agree	shuffle agree
fr	choice	0.9799	0.8750	0.8725
fr	noul	0.9308	0.8100	0.6693
fr	score	0.8957	0.5747	0.3848
da	choice	0.9598	0.8058	0.7852
da	noul	0.9200	0.8067	0.6667
da	score	0.7521	0.5550	0.4300

Table 9: Evidence controls on the elite eval split (n per cell 447–1,218; shuffle excludes 101 donor-overflow rows). Chance is ∼0.25/0.60/0.34 for choice/noul/score (noul majority prior). Controls are scored as agreement with the reference answer under altered evidence, `label_status=evidence_control`.

grounded accuracy, and the battery needs distractors that are indistinguishable without the passage before choice accuracy can be claimed as grounded.

Stricter distractors reduce but do not remove leakage. Rebuilding the choice battery with a strict pool (`--hard-choice`: same-passage swaps and numeric perturbations only, no cross-passage fill; 733 records honestly excluded as untestable, $n = 303$ eval) changes accuracy to 0.914 (fr) / 0.868 (da) and empty-evidence agreement to 0.795 / 0.766. The residual shows that same-passage swaps—answers to *other* questions—remain separable by question–answer coherence alone; fully grounded choice requires confounders to the *same* question. Under shuffled evidence the adapted arm agrees less than the frozen arm (0.722 versus 0.825), consistent with better evidence filtering, matching its SciFact gain.

4.10 Anti-leakage retraining removes the shortcut

The residual leakage could reflect either a capability limit of the frozen backbone or a missing supervision signal. To separate the two we rebuild the noul and score training sets with *evidence-corruption* counterparts: for each clean example we emit an empty-evidence and a shuffled-evidence row whose label is forced to the rejection class (*false* for noul, level 0 for score)—the same supervision the controls measure—subsampled to a fraction ρ of the clean count. Nothing else changes: same head, objective, schedule, and eval files.

Three findings follow. First, the shortcut is a data artifact, not a backbone limit: for noul, corrupted rows collapse every control to the majority prior—balanced accuracy exactly 0.500—at every tested ρ, including ρ=0.0625 where corrupted rows are only 14% of the mix. Score needs a larger dose: controls sit at the prior at $\rho \geq 0.25$, but at ρ=0.0625 shuffled-evidence agreement leaks back to 0.386 (prior 0.334). A modest dose of the right supervision suffices; a token one does not.

Second, the fix *transfers*: on canonical SciFact noul, where the answer must be read off the passage, the retrained heads improve over the frozen release by 23–24 points (0.464→0.693–0.702), while the leaky head sits below its own majority prior. A head trained to exploit option-side leakage cannot use evidence it is given; the retrained heads can, and on the score proxy the ρ=0.25 head matches the leaky release exactly (0.757).

Third, ρ=0.25 is the operating point, not ρ=1. Forcing all corrupted rows to the rejection class shifts the training prior to 89% *false* (noul) and 78% level-0 (score), against eval priors of 67% and 34%, and the 14-point clean-accuracy drop on score at ρ=1 is mostly that threshold shift. At ρ=0.25 score recovers past the release (0.911 versus 0.896 clean) while holding exact-prior controls;

			control agreement		
arm	ρ	clean acc	empty	shuffle	SciFact
noul (elite n=1,218; controls n=1,146; SciFact n=332)					
release `fr`	—	0.931	0.810	0.669	0.464
`al`	1.00	0.896	0.667	0.667	0.693
`al`	0.25	**0.907**	0.667	0.667	**0.702**
`al`	0.0625	0.849	0.667	0.667	0.654
score (elite n=1,218; SciFact proxy n=304)					
release `fr`	—	0.896	0.575	0.385	0.757
`al`	1.00	0.755	0.336	0.334	0.671
`al`	0.25	**0.911**	0.336	0.336	**0.757**
`al`	0.0625	0.906	0.351	0.386	0.655

Table 10: Anti-leakage retraining at corruption fraction ρ (corrupted rows per clean row, split evenly between empty and shuffled). Control cells at the majority prior (0.667 noul, 0.336 score) have balanced accuracy exactly 0.500: the head no longer answers from the option text alone. SciFact cells use the canonical `systemone-v2` conversions and differ from the legacy-file numbers in Table 3; see text.

at ρ=0.0625 it still matches clean but starts leaking again. Note also that the clean-eval accuracies of the leaky arms are themselves inflated by option-side cues shared with the training distribution, so the ρ=0.25 heads are the honest release candidates even where the leaky head retains a nominal edge (noul: 0.907 versus 0.931).

Choice needs a different supervision. The choice head has no rejection class—every row must select one of the candidate answers—so corrupted-evidence rows cannot reuse the label remap. We instead supervise the *distribution*: corrupted rows carry a uniform soft target and are trained with the KL term alone (`--soft-weight 1.0`), expressing "no usable evidence, no preference" without touching the label space. A second mechanism targets evidence-following rather than abstention: provenance-tagged *counterfactual* rows keep question and options but minimally edit the passage so that a documented distractor becomes the supported claim (single-token `number_perturb`-style edits located in the evidence span), with the anti-leakage noul head arbitrating each candidate as a *Proposed answer* verifier (τ=0.6, margin 0.2). Two earlier donor schemes failed for instructive reasons: topical near-passage transplants were rejected by the teacher on all 230 candidates (max p_{true}=0.089), and `cross_passage` provenance splices were likewise all-uniform on 548 candidates—the question does not transfer to a foreign passage, so honest supervision there is exactly the uniform target.

The counterfactual mechanism works where the naive transplant fails: editing the passage rather than replacing it keeps the question applicable, and the teacher confirms the provenance-intended distractor in 50.5% of editable cases (256 of 507), producing hard labels whose gold distribution is position-balanced. The residual gap between `alp25p` and `alp25x` on the controls is the price of commitment: a head trained to follow edited evidence commits more often under destroyed evidence (0.40–0.45 versus 0.26–0.32)—still less than half the release leakage—while recovering clean accuracy. Which trade-off to ship depends on whether the deployment contract prioritizes maximal abstention (`alp25p`) or evidence sensitivity (`alp25x`); the grounded-GPQA probe in Table 6 measures the latter directly.

A caveat on the SciFact cells: Table 3 was produced on the legacy conversion (n=340), whose

			control agreement (chance = 0.25)			
arm	clean	hard	empty	shuf	h.emp	h.shuf
release `fr`	0.980	0.914	0.875	0.872	0.795	0.825
`alpm1` (uniform)	0.964	0.901	0.368	0.479	0.337	0.434
`alp25` (+near)	**0.980**	**0.924**	0.469	0.483	0.459	0.480
`alp25p` (+prov)	0.942	0.871	**0.299**	**0.264**	**0.323**	**0.318**
`alp25x` (+perturb)	0.967	0.898	0.406	0.398	0.432	0.447

Table 11: Choice anti-leakage arms. Empty controls carry uniform soft targets; "+"arms add evidence-edited rows (near/provenance splices that the noul teacher arbitrated, and 256 single-token counterfactual edits confirmed as supporting their provenance distractor). Agreement with the original gold on corrupted evidence is the leakage metric; chance is 0.25. The dose of uniform-supervision rows drives control agreement toward chance, while the 256 confirmed counterfactuals recover 2.5 points of clean accuracy over the uniform-only arm.

options are bare single-token *true/false* candidates scored against varying per-row prompt tails. The canonical `systemone-v2` conversion scores full-phrase options (the verdict token followed by a verbalized verdict sentence) under a fixed question template—the same format the trained heads see in production. On the same claims, the frozen head scores 0.850 in the legacy format but 0.464 under the canonical encoding: the headline number was therefore sensitive to a format the deployed model does not use. For reference, the third-party JEV evaluation scored 0.888 on the same legacy files, so the legacy-format comparison also holds externally. We report both rather than choose; the anti-leakage comparisons in Table 10 are always same-file comparisons.

5 Discussion

What a decision layer buys. A decision layer exposes probabilities without text generation or output parsing. On the elite battery, the frozen heads have ECE 0.0253–0.1221 and choice automation@5% of 0.8070. This retrospective operating curve can inform routing to a slower component, but a deployment threshold must be validated independently under its own data distribution.

Backbone and head must be evaluated together. Readout architecture changes accuracy even with a fixed backbone, while candidate performance also varies with the backbone, context window, and training schedule. The early bw1_sr checkpoint cannot establish a size-dependent knowledge ceiling. The backbone-only diagnostics of Table 4 close the loop on GPQA: no head-free scoring or generation protocol beats the trained head, so the residual gap is backbone knowledge, not readout capacity. Likewise, proximity to a published GPQA score under another inference protocol would not prove that our head extracts all available knowledge. Grounding closes much of the loop in the other direction (Table 6): with oracle evidence the same model reaches 0.70 on GPQA-main, so the readout is not wasting latent competence — but the noisy and diamond cells bound how much a real retriever can buy back.

Candidate selection is not a causal adaptation study. Table 7 supports retaining the frozen heads for this release: the adapted candidates do not satisfy the transfer gate despite some improved elite metrics. It does not establish that domain adaptation necessarily sacrifices generalization, nor that a domain-native backbone would avoid that trade-off. The matched comparison in Table 8 holds head optimization fixed on `systemone-v2` and shows the causal effect is task-dependent:

adaptation transfers to external verification while degrading ordinal scoring. A fuller study should still repeat seeds, restore optimizer state across interruptions, and reserve untouched evaluation data after selection.

Relation to the ecosystem. Concurrent System-One implementations motivate typed decision interfaces. CLM-8B [5] serves the same typed-question contract from a frozen causal encoder whose state/action projection heads are trained contrastively over large state–action corpora, while our implementation explores a masked-diffusion backbone and deterministic option ordering; we additionally train a CLM-style contrastive bi-encoder on our own backbone and data as a matched-architecture control. Backbone diversity also widens the ecosystem's reach: the released CLM-8B heads are tied to the Qwen3-8B encoder, while this work serves the same typed-question contract on a non-Qwen backbone for environments where that dependency is excluded by model-approval or provenance policy. A numerical ranking against other released systems would require matched dataset conversions, splits, and calibration protocols; the empirical tables here report only our own evaluated candidates.

6 Limitations

Backbone pretraining may include benchmark-adjacent text. Heads are trained on one decision distribution and on at most four options; Banking77 performance is poor. Score remains weaker than choice on the elite battery. Results are single runs, without multi-seed uncertainty estimates. Candidate comparisons differ in optimization, and the small backbone is only an early-training control. External benchmarks informed release selection, so they are not an untouched final test set. We report archived evaluation outputs and their hashes, not a new evaluation of the full backbone during release packaging.

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